



Ecology of *Sphaerularia vespae*, a nematode parasite that sterilizes hornets (Vespidae: *Vespa*): host range, geographical distribution and prevalence of parasitism

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Abstract

The insect parasitic nematode, *Sphaerularia vespae*, which causes sterilization of queen hornets, was originally reported in *Vespa simillima* in Hokkaido and more recently from *V. mandarinia* and *V. ducalis* in Kyoto, Japan. However, its parasitism in the field is not fully understood. In this study, six native hornet species (*V. simillima*, *V. mandarinia*, *V. analis*, *V. ducalis*, *V. crabro* and *V. dybowskii*) were collected from numerous sites in the Japanese main islands to investigate the host ranges, geographical distributions, and annual and seasonal changes in prevalence of *S. vespae*. We determined the hosts of *S. vespae* to be the queens of five of the six hornet species except *V. analis*. Workers and males were not hosts. Finding *S. vespae* from cool to warm regions indicates the likelihood that this nematode's distribution mirrors that of its host hornets. Parasitism of *S. vespae* at two study sites where consecutive 10-year observations were conducted persisted in the dominant hornet species, which was *V. simillima* in a cooler region and *V. mandarinia* in a warmer region. At these sites, no clear relation was apparent between the prevalence of *S. vespae* and the number of host hornets captured. A rising prevalence during the season was observed in *V. simillima* but not in *V. mandarinia*, possibly caused by the species-dependent characteristics of where the colony was founded. Persistence of *S. vespae* parasitism in host hornets and the potential of *S. vespae* as a biological control agent against exotic hornets were discussed.

Keywords Biological control · Invasive species · Natural enemy · Sterilization · Wasp

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Introduction

Vespine wasps (the hornets and yellowjackets) play important roles as predators in various environments (Edwards 1980; Matsuura and Yamane 1990). They act as beneficial natural enemies of agricultural or forest pests (Matsuura and Yamane 1990). However, of the vespines, hornets, or *Vespa* Linnaeus species, are of greater concern than those of *Vespa* Thomson or *Dolichovespula* Rohwer, as they tend to be serious raiders of beehives (Matsuura and Sakagami 1973). They also frequently establish nests in residential areas (Makino et al. 1981; Matsuura and Yamane 1990), which increases the risk of human stings. Since hornets have both positive and negative impacts on humans, a better understanding of their natural enemies is needed, as their populations are possibly affected by natural enemies, including parasites (Schmid-Hempel 1998).

The parasitic nematode *Sphaerularia vespae* Kanzaki et al. was first found in Hokkaido from the common hornet *V.*

simillima Smith (Sayama et al. 2007; Kanzaki et al. 2007) as the second discovered species of the genus *Sphaerularia* Dufour according to Siddiqi (2000). Another *Sphaerularia* species, *S. bombi* Dufour, is a well-known parasite of bumblebees (*Bombus* Latreille spp.) because of its parasitic sterilization of host gynes, or potential queens (Macfarlane et al. 1995; McCorquodale et al. 1998; Poinar and van der Laan 1972; Schmid-Hempel 1998) and the unique morphology of the hypertrophied egg-producing sac called the “uterium” (Siddiqi 2000), which is the female reproductive system everted from the main body of the nematode. *Sphaerularia vespae* also sterilizes its host overwintered queens (gynes) and has similar morphological characters to those of *S. bombi* (Sayama et al. 2007).

The prevalence (percentage parasitism) of *S. vespae* in overwintered queens of *V. simillima* collected with bait traps is very high, at around 70% (Sayama et al. 2007; 2013). The nematode is therefore a potential agent that might affect the population of hornets. However, our present knowledge of the parasite is limited to its prevalence in only one host species (*V. simillima*) in a single locality, Sapporo in northern Japan (Sayama et al. 2007) and to elements of its method of infection (Sayama et al. 2013). Although *S. vespae* was recently reported in *V. mandarinia* and *V. ducalis* Smith in Kyoto, central Japan (Kanzaki et al. 2023), this information is restricted to one study site for one season. Its host range in other *Vespa* species, geographical distribution, and long-term variations in prevalence are largely unknown.

Six native species of *Vespa* are distributed in the four main islands of Japan (Hokkaido, Honshu, Shikoku and Kyushu): *V. simillima*, *V. mandarinia*, *V. analis* Fabricius, *V. ducalis*, *V. crabro* Linnaeus and *V. dybowskii* André. Except for the social parasite *V. dybowskii*, which usurps established nests of *V. crabro* or *V. simillima* (Matsuura 1991), they have similar annual life cycles (Matsuura and Yamane 1990). Overwintered queens establish their colonies in spring, and workers are produced from early to middle summer. Males and new queens are produced from late summer to autumn. After mating, new queens overwinter in environments such as in rotten wood, and establish colonies in spring next year. The queens of *V. simillima* parasitized by *S. vespae* frequently visit potential overwintering sites and release juvenile nematodes (Sayama et al. 2013). The released nematode juveniles remain there and grow into adults (males and infective females). After mating, the infective females of *S. vespae* infest overwintering queens of *V. simillima*. The infective females develop uteria and lay eggs in the body cavity of overwintered queens.

In this study, more than 10,000 overwintering and overwintered queen, worker, and male hornets were collected across the Japanese main islands to clarify the host range, geographical distribution and geographical variations in

the prevalence of *S. vespae*. Surveys were conducted for 10 years at two locations to investigate their long-term prevalence. Based on these results, we discuss how parasitism of *S. vespae* persists in host hornets and the potential of *S. vespae* as a biological control agent against exotic hornets.

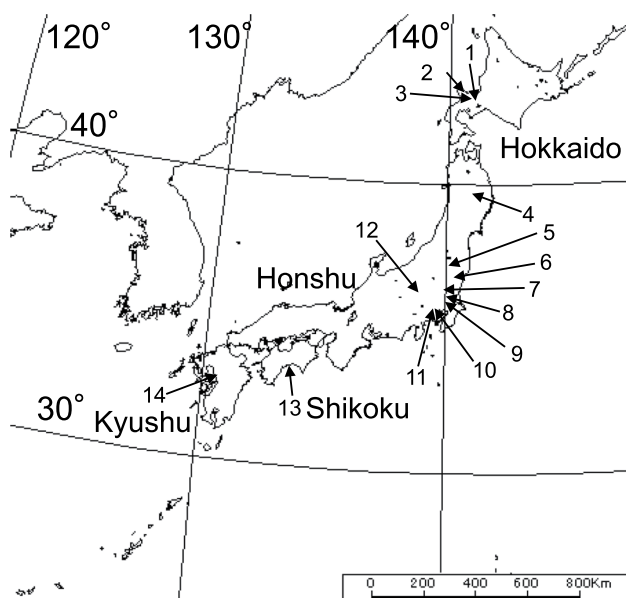
Materials and methods

The six native species of hornets were collected from 14 localities in Japan (Table 1 and Fig. 1). These localities cover the four main islands (Hokkaido, Honshu, Shikoku and Kyushu), which range climatically from the subarctic to the warm temperate zone. The collection was conducted in forest parks, experimental forests and secondary forests, all of which were mainly composed of broad-leaved trees and in conifer plantations. Basic collection methods of active hornets (overwintered queens, workers and males) were as follows. Active hornets were trapped using 2000-ml clear plastic bottles with a small open window in the upper part and a mixture of clear liquor (distilled spirit) and orange juice as bait in the bottle (Makino and Sayama 2005). This trap is considered useful for monitoring the diversity and abundance of vespine species (Makino 2022). The traps were tied to trees at a height of approximately 1.3 m. Renewal of bait and collection of trapped insects were carried out every two weeks. To study the geographical variations in the prevalence of *S. vespae* in *V. simillima*, overwintered queens were collected using these traps in spring to summer in nine of the 14 localities (Table 1). At some collection sites, *V. simillima* was singled out for surveying from the various collected hornet species. The number of traps ranged from three to 30 depending on the locality (Table 1).

In Sapporo, Karuizawa, Kumamoto and Kochi, to reveal the host range of *S. vespae*, captured overwintered queens of every *Vespa* species were dissected. At the former three collection sites, hornets were also collected during their active seasons (from spring to autumn), and workers and males were also dissected. Multi-year surveillance was conducted at these three collection sites. The details of surveillance at these collection sites are as follows. In Sapporo, bait trapping was conducted from 2005 to 2014 at the Forestry and Forest Products Research Institute's (FFPRI's) Hitsujigaoka Experimental Forest. Ten traps were attached to trees along walking trails at 50-m intervals. In 2007 and 2008 queens, workers and males of every *Vespa* spp. were dissected and in the other years (2005 and 2006 and 2009–2014), only queens of *V. simillima* were dissected. Parts of results in 2005 and 2008 have already been reported, which include the first record of *S. vespae* from the hornets (Sayama et al. 2007) and evidence that *S. vespae* manipulates host behavior (Sayama et al. 2013). In Karuizawa, three bait

Table 1 Locality, collection method and collection period where *Vespa* spp. were collected

No. in Fig. 1	Locality	Latitude (N)	Longitude (E)	Altitude (m)	Climate ^a	Environment	Collection source	Collection period yy/mm
1	Ebetsu	43°02′	141°31′	40	SA	Forest park	Hibernacula	2007/10–11
2	Otaru	43°12′	140°58′	80	SA	Forest park	Hibernacula	2007/11
3	Sapporo	42°59′	141°23′	150	SA	Experimental forest	Bait trap (10) ^b Hibernacula	2005/05– 2014/10 2007/04– 2009/04 ^c
4	Morioka	39°46′	141°08′	190	SA	Experimental forest	Hibernacula	2008/11
5	Sukagawa	37°17′	140°22′	270	T	Forest Park	Hibernacula	2009/01
6	Kitaibaraki	36°56′	140°35′	580–800	T	Secondary forest	Bait trap (30)	2002/05–06
7	Shirasato	36°27′	140°16′	250	T	Conifer plantation	Bait trap (5)	2008/05–06
8	Ishioka	36°13′	140°07′	450	T	Conifer plantation	Bait trap (5)	2008/05–06
9	Tsuchiura	36°04′	140°09′	20	T	Secondary forest	Hibernacula	2010/04
10	Tama	35°38′	139°27′	90	T	Experimental forest	Bait trap (20)	2008/06–07
11	Hachioji	35°38′	139°16′	200	T	Experimental forest	Bait trap (30)	2008/05–06
12	Karuizawa	36°20′	138°31′	1010	SA	Conifer plantation	Bait trap (3)	2018/05– 2019/12
13	Kochi	33°32′	133°28′	70	WT	Experimental forest	Bait trap (15 or 7) ^d	2014/04–05
14	Kumamoto	32°49′	130°44′	50, 100, 150 ^e	WT	Experimental forest	Bait trap (6)	2009/05– 2018/12

^aSA subarctic; T temperate; WT warm temperate^bNumber of traps in parentheses^c2007/04 and 11, 2008/04, 10 and 11, 2009/04^d15 traps to 9 May and after that, 7 traps^eTraps were placed at the base, halfway up, and at the top of a low mountain**Fig. 1** Localities in Japan at which *Vespa* spp. were collected. Numbers are complementary with Table 1, in which information on individual localities is given

traps were set in 2018 and 2019 in the garden of a private home at 20-m intervals. In Kumamoto, traps were placed in the FFPRI's Tatsudayama Experimental Forest from 2009 to 2018. Three pairs of traps each at 400-m intervals were

placed at the foot, halfway up and at the top of Mt. Tatsudayama (152 m a.s.l.). Each pair of traps was composed of two traps: one used the standard bait (mixture of orange juice and liquor) and the other used a mixture of a different brand of clear liquor and orange juice (2013–2016) or one third of diluted honey with tap water (volume by volume) (2009–2012 and 2017–2018). Although the effect of differences in bait composition on catch cannot be completely ruled out, no changes consistent with bait components were observed in the number of trapped hornets (see Fig. 3), and it is unlikely that usage of different baits affected the conclusions of the present study. Parts of results from 2012 to 2018 have already been published (Kosaka et al. 2021) in which a parasitic mite of *Vespa* spp. was reported. Annual changes in the prevalence of *S. vespae* in overwintered queens were studied in Sapporo in *V. simillima* populations and in Kumamoto in *V. mandarinia* populations over a period of ten years. To examine whether changes in prevalence affect changes in capture numbers, we conducted a correlation analysis between the rate of change in the prevalence from the previous to current year (P_t/P_{t-1}) and the rate of change in the number of captured hosts from the current to following year (H_{t+1}/H_t), where P represents prevalence, H is the number of captures, and t denotes the current year. However, the data from Kumamoto in 2013 were excluded from

the analysis because the capture number of that year was extremely and abnormally low. Monthly seasonal changes in prevalence through the study years in Karuizawa and Kumamoto and of parts of study years in Sapporo (2007–2009) were also observed. In these observations data with sample sizes (number of queens collected) smaller than five were not used in the analysis. In Sapporo, the changes were observed continuously using prevalence in overwintering queens. In Kochi, 15 bait traps were placed in the FFPRI's Experimental Forest at 50-m intervals from 25 March to 9 May and seven traps at 100-m intervals to 19 May 2014. Parts of results have already been published (Sato et al. 2015) in which the local species composition of *Vespa* spp. was described.

Collection of overwintering queens in hibernation was made in six localities in winter (Table 1): we explored decayed wood (stumps or logs) in small forests or coppices, and gently removed bark and wood using a small pickaxe to find queens in hibernacula (Matsuura and Yamane 1990).

Hornet specimens collected in this study approximately amounted to 10,000. These collections, however, were made in widely dispersed parts of Japan over several years. In addition, there was no recognizable negative trend in the number of collected hornets in the study sites where we continuously trapped them for several years (see results). Then we can reasonably think that our collections did not affect hornet populations.

Collected specimens were dissected under a stereomicroscope to check for parasitism by *S. vespae*, and if found, the developmental states of the parasites (with or without oviposition and growth of juvenile nematodes) were recorded.

An additional observation was conducted in Sapporo. The development of uteria in overwintering queens of *V. simillima* (collected on 20 April 2007) and *V. analis* (27 April 2007) were qualitatively compared using photos.

During the study, we confirmed that all the *Sphaerularia* nematodes from the hornets were *S. vespae*. Juvenile nematodes from hornets have a flat head and one or more constrictions in the tail: these morphological characteristics are typical of *S. vespae* (Kanzaki et al. 2007), whereas juvenile *S. bombi* has a round head and no constrictions in the tail (Poinar and van der Laan 1972). The molecular profiles of the nematodes from *V. analis* and *V. dybowskii* were analyzed using the methods of Kanzaki et al. (2007), and the sequences were found to be identical to those from *S. vespae* (AB300595 and AB300596). For the nematodes from *V. mandarinia* and *V. ducalis*, their identity to *S. vespae* was shown by Kanzaki et al. (2023). Although the molecular profile of the nematode from *V. crabro* has not been analyzed yet, its identity to *S. vespae* is morphologically confirmed as stated above.

Results

Host range and geographical distribution

Sphaerularia vespae was detected in overwintered and/or overwintering queens of all of the six *Vespa* species from the northern to southern parts and the low to high altitudes of Japan (Table 2). In *V. analis*, however, *S. vespae* was detected only from the overwintering queens in Sapporo but not from the overwintered queens from any localities including Sapporo. The uterium of *S. vespae* in overwintering *V. analis* was much smaller than those in *V. simillima*, which were collected at the same locality and at almost same time (Fig. 2). The uterium length in *V. analis* was about 2 mm, much smaller than in *V. simillima*, in which it generally exceeded 5 mm. Egg-laying uteria and hatched developing juvenile nematodes were found in all *Vespa* spp. after late spring except for *V. analis*. Dissections of more than 7000 workers and 700 males revealed no *S. vespae* in workers and males of *Vespa* spp. (Table 3).

Sphaerularia vespae was found in overwintered queens of *V. simillima* at five of nine collection sites installed with bait traps (Table 2). In Kitaibaraki and Kumamoto the prevalence was less than 10%, whereas in the other three localities it exceeded 40% (Table 2). Of the overwintered queens in Sapporo, Karuizawa and Kumamoto, the highest prevalence of *S. vespae* was observed in the most-commonly captured species at each collection site: *V. simillima* in Sapporo and Karuizawa and *V. mandarinia* in Kumamoto (Table 2). In other species, *S. vespae* was detected in only one individual queen at these collection sites: *V. dybowskii* in Sapporo and *V. simillima*, *V. crabro* and *V. ducalis* in Kumamoto. In Karuizawa, *S. vespae* was detected in only *V. simillima*. *Sphaerularia vespae* was not detected in Kochi in five hornet species where *V. analis* was dominant. Other than in Kochi, *S. vespae* was not found at three collection sites, Shirotsato, Ishioka and Hachioji, where only *V. simillima* was surveyed (Table 2).

In overwintering *V. simillima* queens, *S. vespae* was found at five of six collection sites. In Tsuchiura, where *S. vespae* was not found in *V. simillima*, the nematode was detected in *V. mandarinia* (Table 2).

Annual and seasonal changes in the prevalence of the nematode

Figure 3 shows the annual changes over 10 years in the capture number of overwintered queens of *V. simillima* in Sapporo and *V. mandarinia* in Kumamoto and the prevalence of *S. vespae* in these queens. The annual numbers of hosts collected by traps fluctuated widely. The minimum and maximum numbers were 11 and 122 in *V. simillima*, and three

Table 2 Prevalence (%) of *Sphaerularia vespae* in queens of *Vespa* spp. from different localities in Japan. Numbers of hosts examined are given in parentheses

Locality	<i>V. simillima</i>	<i>V. mandarinia</i>	<i>V. analis</i>	<i>V. ducalis</i>	<i>V. crabro</i>	<i>V. dybowskii</i>
(A) Overwintered queens collected with bait traps						
3 Sapporo ^a	60.7 (145)	0 (14)	0 (63)	ND ^b	0 (43)	2.1 (48)
6 Kitaibaraki	7.3 (41)	* ^c	*	*	*	*
7 Shirosato	0 (23)	*	*	*	*	*
8 Ishioka	0 (41)	*	*	*	*	*
10 Tama	66.7 (3)	*	*	*	*	*
11 Hachioji	0 (29)	*	*	*	*	*
12 Karuizawa	43.8 (105)	0 (14)	0 (34)	0 (4)	0 (3)	0 (1)
13 Kochi	0 (7)	0 (43)	0 (101)	0 (1)	0 (4)	ND
14 Kumamoto	7.7 (13)	18.0 (1197)	0 (616)	0.2 (411)	1.0 (100)	ND
(B) Overwintering queens from hibernacula						
1 Ebetsu	28.6 (7)	^d	0 (18)	ND	—	—
2 Otaru	60 (5)	—	0 (1)	ND	—	—
3 Sapporo ^e	35.2 (71)	—	5.2 (116)	ND	—	—
4 Morioka	25 (4)	—	0 (1)	—	—	—
5 Sukagawa	25 (4)	—	0 (2)	—	—	—
9 Tsuchiura	0 (4)	33.3 (3)	0 (2)	0 (1)	—	—

^aResults for 2007 and 2008

^bHornets of this species not distributed

^cNot examined for parasitism of *S. vespae*

^dHornets not found

^eResults for April 2007 to April 2009

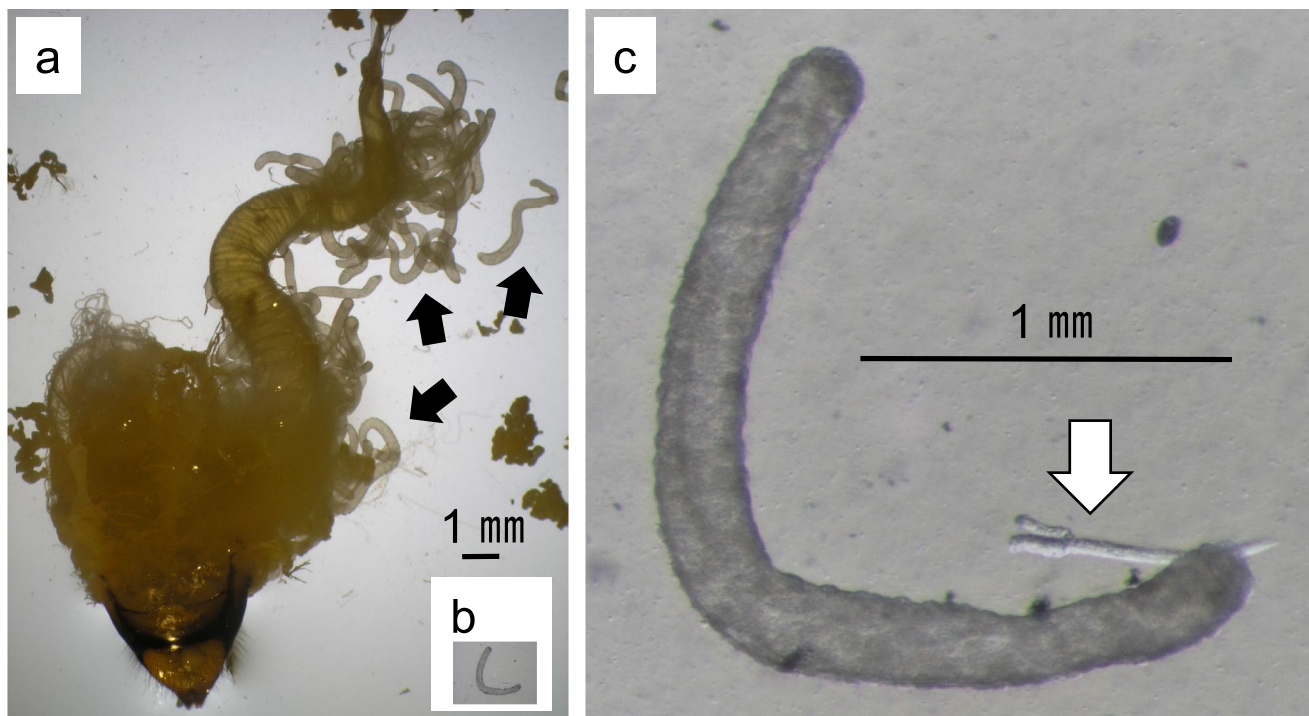


Fig. 2 Examples of *Sphaerularia vespae* uteria in overwintering queens of *Vespa* spp. collected in April 2007. **a** In *V. simillima*, a total of 125 individual uteria (black arrows) were observed in this case. The body length of most individuals exceeded 5 mm. **b** In *V. analis*, one

smaller uterium with body length of ca. 2 mm was observed (same scale as in **a**). **c** Magnification of (**b**). The main body of a nematode (white arrow) was found in this case

Table 3 Number of workers and males of *Vespa* spp. captured with bait traps and dissected^{a,b}

Species	Locality	Workers	Males
<i>V. simillima</i>	Sapporo	101	7
	Karuizawa	244 ^c	62
	Kumamoto	149	10
<i>V. mandarinia</i>	Sapporo	83	31
	Karuizawa	51	4
	Kumamoto	3626	180
<i>V. analis</i>	Sapporo	400	101
	Karuizawa	42	4
	Kumamoto	1508	102
<i>V. ducalis</i>	Karuizawa	8	0
	Kumamoto	122	50
<i>V. crabro</i>	Sapporo	248	69
	Karuizawa	1	0
	Kumamoto	627	82
<i>V. dybowskii</i>	Sapporo	172	7
	Karuizawa	2 ^d	0

^a*Sphaerularia vespae* was not found in any of the workers or males

^bCollection years are the same as in Table 1 except for *V. simillima* (2007–2008) in Sapporo

^cOne individual was not dissected for specimen preparation

^dTwo individuals were not dissected for specimen preparation

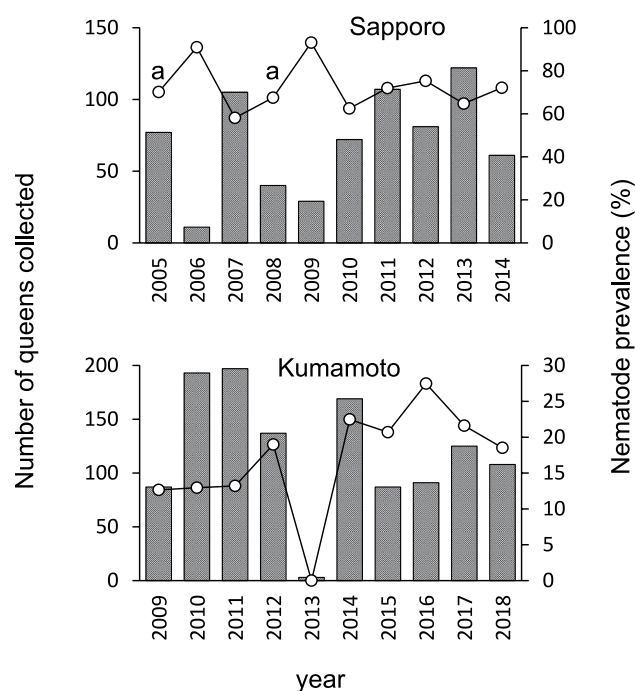


Fig. 3 The prevalence of *Sphaerularia vespae* in *Vespa simillima* in Sapporo and in *V. mandarinia* in Kumamoto is given as empty dots, connected by lines over the multiple years. ^aValues for prevalence in 2005 and 2008 in Sapporo are cited from Sayama et al. (2007) and Sayama et al. (2013), respectively

and 197 in *V. mandarinia*. *Sphaerularia vespae* was detected every year in *V. simillima*. The prevalence of *S. vespae* in *V. simillima* ranged from 58% to 93% and the mean prevalence over 10 years was 69%. *Sphaerularia vespae* was detected every year in *V. mandarinia*, except in 2013 when the annual catch of queens was the smallest during the study years. Annual prevalence of *S. vespae* in *V. mandarinia* was 13%–27%, except for 2013, and the mean prevalence over 10 years was 18% including the results for 2013. No clear relationship was found between the rate of change in the prevalence from the previous to current year and the rate of change in the number of captured hosts from the current to following year in either *V. simillima* (correlation coefficient = 5.469, $r^2 = 0.245$ and $p = 0.213$) or *V. mandarinia* (correlation coefficient = -0.655, $r^2 = 0.126$ and $p = 0.491$). Seasonal changes in the number of queens collected and in the prevalence of *S. vespae* are shown in Fig. 4 for *V. simillima* in Sapporo (2007–2009) and Karuizawa (2018–2019), and for *V. mandarinia* in Kumamoto (2009–2018). The prevalence in *V. simillima* tended to rise later in the trapping season in both Sapporo and Karuizawa. In Sapporo the prevalence in the overwintering queens was 32%–50%, lower than that in overwintered queens except for May 2007. On the other hand, its prevalence in *V. mandarinia* tended not to rise later in the trapping seasons and not to show specific seasonal changes.

Discussion

Sphaerularia vespae was found in all the six native species of queens of *Vespa* spp. However, in overwintered queens of *V. analis*, the dissections of over 800 queens revealed no egg-laying uterium of *S. vespae*. Additionally, the uterium of *S. vespae* in an overwintering queen of *V. analis* was much smaller than those in *V. simillima*. These results suggest that even if *S. vespae* is able to infest overwintering queens of *V. analis*, it cannot develop an egg-laying uterium in the overwintered queens. *Vespa analis* queens may therefore have some resistance to the development of *S. vespae*, though its mechanism is unknown at present. We conclude that *V. analis* is not a host for *S. vespae*. The overwintering sites of *V. simillima* where the infection of *S. vespae* occurs (Sayama et al. 2013) coincide with those of *V. analis*, such as in rotten wood where overwintering queens of these two species are frequently found (Matsuura and Yamane 1990). It should be possible for *S. vespae* derived from queens of *V. simillima* to infest overwintering new queens of *V. analis*. This infection, if it takes place, is likely an erratic parasitism.

Egg-laying uteria and developing juvenile nematodes were observed in five native species of queens of *Vespa* spp. except *V. analis*. No *S. vespae* were found through the

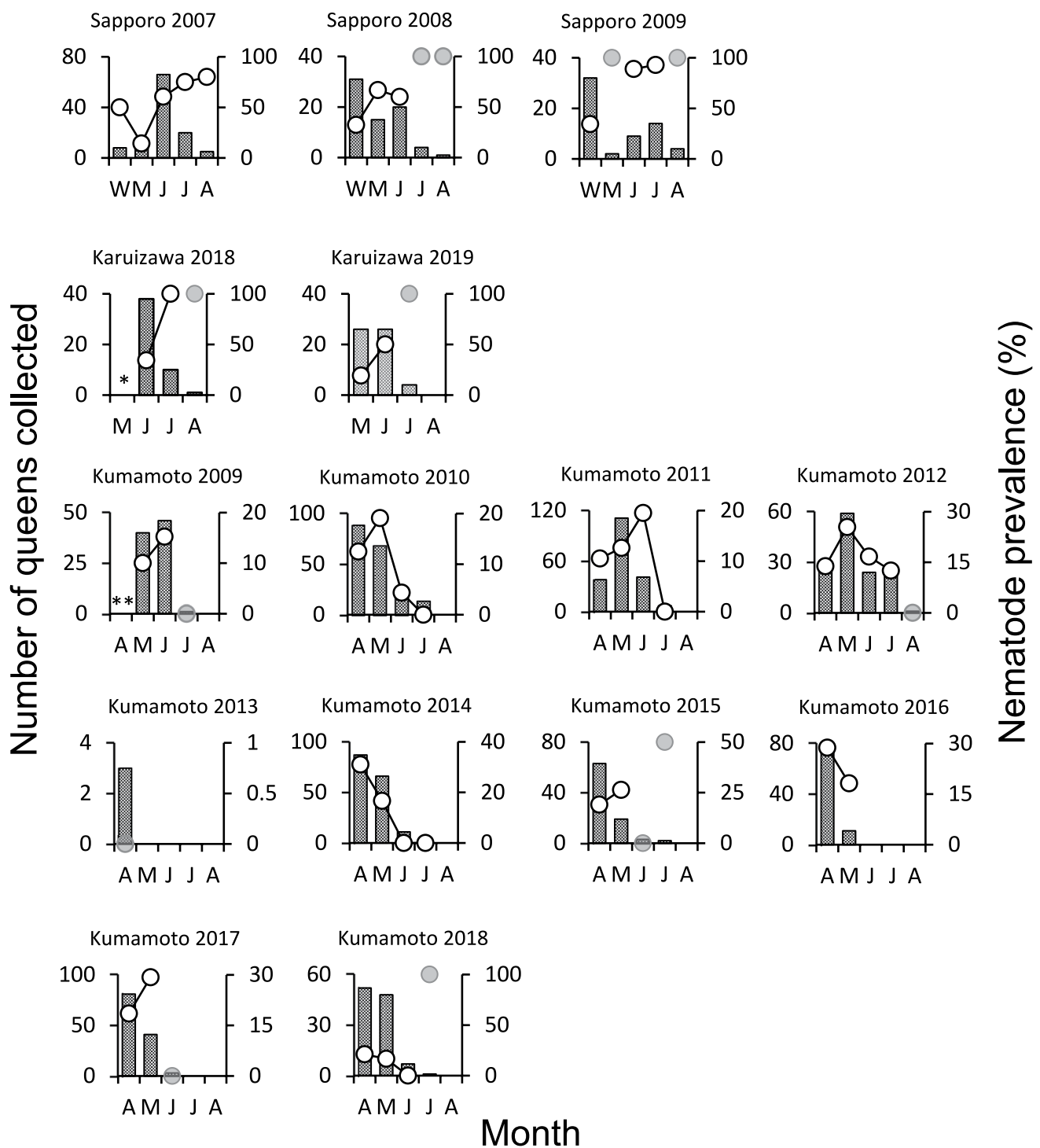


Fig. 4 The prevalence of *Sphaerularia vespae* in *Vespa simillima* in Sapporo and Karuizawa and in *V. mandarinia* in Kumamoto is given as empty dots, connected by lines over the multiple sampling times over the season (W, A, M, J, J, A: winter, April, May, June, July, August). Data with sample sizes (number of queens collected given as bars)

smaller than five are indicated by the faint dots and were not used in the analysis. “Winter” for Sapporo indicates data for overwintering queens. *Data combined with June. **No data available. Data of Sapporo 2008 were modified from Sayama et al. (2013), which presented seasonal changes in the number of parasitized queens

dissections of 7000 worker and 700 male hornets. From these observations, the hosts of *S. vespae* among the six hornet species native to the Japanese main islands are queens of *V. simillima*, *V. mandarinia*, *V. ducalis*, *V. crabro* and *V. dybowskii*.

The phylogenetic analysis of *Vespa* spp. has been conducted (Perrard et al. 2013), revealing that there are three main clades of hornet species and hosts of *S. vespae* are widely distributed across these clades. Notably, *V. simillima* and *V. analis* belong to the same clade, and *V. analis* represents the earliest-diverging species in the nine species in this clade. It would be intriguing if speciation of *V. analis* were associated with the acquisition of the resistance against the development of *S. vespae*.

Sphaerularia vespae was found from northern to southern parts and low to high altitudes of Japan, ranging from the subarctic to the warm temperate zones. This indicates that *S. vespae* is most likely distributed wherever host hornet species are present. *Sphaerularia vespae* was recently found in Kyoto, in the western part of Honshu and not covered in this study, in *V. mandarinia* and *V. ducalis* (Kanzaki et al. 2023). Their report supports the conclusion of the present study concerning its geographical distribution.

Sphaerularia vespae was detected almost every year from *Vespa* spp. whose capture number by bait traps was the highest in each study area: *V. simillima* in Sapporo and Karuizawa, which are cool climes with high latitude or altitude, and *V. mandarinia* in Kumamoto (except 2013), a warm clime with low latitude; but rarely in other hornet species. Makino (2022) concluded bait traps (attraction traps) to be a more effective collection method for revealing the diversity and abundance of *Vespa* spp. than the Malaise trap (a flight interception and passive collection trap). Hornet species whose capture number by bait traps in this study was highest should be dominant species with the highest density in the study areas. It makes logical sense for *S. vespae* to maintain their population in the dominant *Vespa* species, since the infection of disease or parasitic organisms is a function of host density (Anderson and May 1978, 1979). It appears likely that *S. vespae* uses dominant *Vespa* species as their main host to maintain its population but occasionally parasitizes other non-dominant species as minor hosts.

Infection of *S. vespae* seems to continue in the specific hornet species because infection occurs at species-specific overwintering sites. Nevertheless, parasitism of *S. vespae* did not continue in the non-dominant host hornet species in this study. It appears that the infectivity of *S. vespae* is not strong enough to maintain its population if it parasitizes non-dominant hosts. The reason that non-dominant host hornets were infected with *S. vespae* is probably the use of the same overwintering sites, which are the infection points of *S. vespae*, by different hornet species. These

overwintering sites are in the soil for all six native hornet species (Matsuura and Yamane 1990) and in the rotten wood as well for five species except for *V. crabro* (Matsuura and Yamane 1990; Sayama, pers. obs. for *V. dybowskii* and this study for *V. mandarinia*). It appears that *S. vespae* derived from dominant and main host hornets rarely infests non-dominant host hornets.

If it is true that *S. vespae* maintains its population only in dominant host hornet species, this might explain why *S. vespae* was not detected in Kochi, where the dominant species was the non-host species, *V. analis*. *Sphaerularia vespae* might not maintain its population in areas where *V. analis* is dominant even if a non-dominant host hornet species harboring *S. vespae* were to invade such areas. In the other three localities, Shirosato, Ishioka and Hachioji, where *V. simillima* was selectively collected and *S. vespae* was not detected, *V. analis* may be dominant or *S. vespae* may use other dominant hornet species. It will be interesting, in further studies, to elucidate the parasitism of *S. vespae* in areas where the dominant hornet species is *V. analis* or the abundances are antagonistic among hornet species.

Sphaerularia vespae infects overwintering queen hornets. Its prevalence should therefore remain constant during the season of the same hornet species. Nonetheless, the prevalence of *S. vespae* in *V. simillima* increased later in the season. Queens of *Vespa* species parasitized by *S. vespae* appear to continue wandering in the field during the day, since they cannot establish a colony due to having been sterilized (Sayama et al. 2007). On the other hand, healthy queens that found their colonies leave their nests only when necessary, probably spending much less time in the field than do parasitized queens, which might explain the rise in prevalence of *S. vespae* in *V. simillima* during the season. This study gave the same result as a previous study in Sapporo (Sayama et al. 2007), and also indicates that the annual prevalence in overwintered queens of *V. simillima* in Sapporo, at 58%–93%, is an overestimation. The real prevalence is probably 32%–50% as was observed in overwintering queens. In *V. mandarinia*, specific seasonal changes in the prevalence of *S. vespae* were not found in overwintered queens. Queens of *V. mandarinia* establish their nests in an enclosed space, such as in the soil or the tree hollow, whereas *V. simillima* establishes its nests both in open spaces like under tree branches or even under man-made structures such as the eaves of houses or bridges, and in enclosed spaces (Yamauchi 2009). It may be more difficult for *V. mandarinia* than *V. simillima* to find a place that is suitable for establishing a colony. As a result, healthy queens may continue their activity in the field for longer; the prevalence of *S. vespae* in overwintered queens of *V. mandarinia* shows no specific seasonal changes. The actual prevalence of *S. vespae* in *V. mandarinia* in Kumamoto is difficult to

estimate because the prevalence of overwintering queens is not known. Given the lack of specific seasonal changes in the prevalence of *S. vespae* in overwintered queens of *V. mandarinia*, the real prevalence may be guessed to be almost the same as the annual prevalence in overwintered queens for 10 years of this study, i.e., at around 20%.

Sphaerularia vespae will affect the host abundance to a certain extent because it sterilizes queens as hosts. This has been of great concern from the beginning of the study. *Sphaerularia vespae* probably does not affect populations of the minor hosts because its prevalence in these species was very low. On the other hand, the prevalence of *S. vespae* in the main hosts was about 20% in *V. mandarinia* in Kumamoto and 32%–50% in *V. simillima* in Sapporo, but it was difficult to find a specific relation between the changes in 10-year annual capture numbers of the two host species and the prevalence of *S. vespae*. For example, no clear relationship was found between the rate of change in the prevalence from the previous to current year and the rate of change in the number of captured hosts from the current to following year in either *V. simillima* or *V. mandarinia*. The effect of *S. vespae* on the population dynamics of the main hosts is likely smaller than some other effects such as success or failure to establish a colony by healthy queens. This may be one reason why *S. vespae* and its host hornets can coexist.

Recently the yellow-legged hornet *V. velutina* Lepeletier invaded to Europe (Monceau et al. 2014), Korea (Kim et al. 2006) and Japan (Takeuchi et al. 2017). The natural enemies of this invasive hornet have been intensively explored: these include entomopathogenic fungi (Poidatz et al. 2018), a mermithid nematode (Villemant et al. 2015) and a conopid parasitoid fly (Darrouzet et al. 2015). It would be worthwhile to examine potential parasitism of *S. vespae* on *V. velutina* in Japan. Parasitism might occur because *V. velutina* is closely related to *V. simillima* (Perrard et al. 2013). If so, *S. vespae* could be considered a potential biological control agent of *V. velutina* in Japan as *S. vespae* is native while *V. velutina* is invasive to Japan. It is necessary, however, to carefully explore possible side effects on the environments as well as on the other host species before the usage.

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Declarations

Conflict of interest The authors declare no conflicts of interest.

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